

Licenciatura em Engenharia Informática

LEI-ISEP 5th Semester Integrative Project

PART II – System Specification

Version v1.0

This document presents a *Request for Proposal* (RFP) issued by **LaPrizza**, a modern pizza restaurant seeking a technological partner capable of designing and developing a new information system to support and improve its business and operational processes.

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II.1 Preamble

In the academic year 2026/2027, the fifth semester (i.e., the 3rd year, 1st semester) of the Bachelor's Degree in Informatics Engineering (**LEI**) at the Instituto Superior de Engenharia do Porto (**ISEP**) adopts a teaching-learning approach centred on the development of a single integrative project. This project promotes the integration and practical application of the knowledge, skills, and competencies acquired across all course units (UCs) taught during the semester: Administração de Sistemas (**ASIST**); Arquitetura de Sistemas (**ARQSI**); Inteligência Artificial (**IART**); Laboratório e Projeto V (**LAPR5**); and Sistemas Gráficos e Interação (**SGRAI**).

The project, common to all course units, consists of developing the information system specified in this document. Its development must follow the general principles, procedures, and operational guidelines established in the companion document entitled *PART I – Context and Operation*.

II.2 Context

This document simulates a *Request for Proposal* (RFP) issued by **LaPrizza**, a modern pizza restaurant seeking a technological partner capable of designing and developing a new information system to support and improve its business and operational processes.

As part of the selection process, interested Information Technology (IT) companies are invited to develop an initial, simplified prototype of the proposed system over a period of 14 weeks, organized into three sprints. Within the context of this academic exercise, each student team represents a different IT company and is expected to prepare a proposal and develop a prototype as if participating in a real-world competitive selection process.

The resulting prototype will serve as a *proof of concept*, demonstrating the integration of multiple technologies, the feasibility of the proposed architecture, and the team's ability to address the functional and non-functional requirements defined in this RFP. The selection criteria extend beyond purely technical considerations and also encompass professionalism, adaptability, quality of delivery, and a sound understanding of the restaurant business and its operational context.

To reinforce this business dimension, each team must select and study a real company operating in the restaurant sector. This analysis should provide a deeper understanding of the sector's business and operational practices, as well as relevant organizational and management concepts. The knowledge gained from this study should contribute to the development of a more coherent and well-founded proposal aligned with **LaPrizza's** business needs.

II.3 System Description

The system envisioned by **LaPrizza** aims to support the operation and management of a modern pizza restaurant through an integrated information system covering areas such as customer management, product and allergen management, order processing, stock control, pizza configuration, food preparation, interactive 3D visualization, intelligent planning and scheduling, and systems administration.

Operating a pizza restaurant involves several interdependent business concerns that need to be managed in a coordinated manner. Products may serve different commercial and operational purposes: some are offered directly to customers, while others may be used as components in the configuration and preparation of pizzas. Some products may also be subject to stock control, making their operational availability dependent on the quantities currently available. Allergen information associated with products is relevant when presenting and configuring food items, while confirmed customer orders generate preparation work that must be coordinated using the available kitchen resources. Together, these concerns provide the business context for the different capabilities explored throughout the project.

The proposed solution is envisioned as a modular, web-based information system organized around four major technological areas: the **Web Application**, **3D Visualization and Interaction**, **Artificial Intelligence**, and **Systems Administration and Infrastructure**. These areas are expected to cooperate through clearly defined interfaces and APIs while remaining sufficiently independent to support modular development, deployment, maintenance, and evolution.

The system described in the following sections represents a deliberate simplification of a real-world restaurant information system. The descriptions establish the **business and technological context** in which the project is developed and identify capabilities that may progressively become part of the prototype. They should not, by themselves, be interpreted as requiring the implementation of every capability described.

Scope of the System Description

This section provides the **business context and overall vision** for the **LaPrizza** information system. The functionality required in each development iteration is established by the corresponding **Sprint Backlog and User Stories**.

Consequently, a capability described in this section is not necessarily required to be implemented unless it is explicitly included in the applicable Sprint Backlog or otherwise identified as a project requirement.

II.3.1 Business Overview

LaPrizza is a modern pizza restaurant that prepares and sells pizzas, beverages, and other food products to its customers. Its operation involves managing customers and products, processing customer orders, configuring pizzas, monitoring the availability of relevant products, and coordinating the resources and activities required to prepare and serve those orders.

Different types of users may interact with the information system according to their responsibilities and permitted operations. Customers may interact directly with some of the services provided by **LaPrizza**, while restaurant staff may use the system to support the management and operation of the business. Users can therefore be identified within the system and assigned appropriate roles that determine the operations they are authorized to perform. Customers may also be allowed to register themselves and, where applicable, record their preferences and consent regarding marketing communications and other optional uses of their personal data.

II.3.1.1 Customers and Food Intolerances

LaPrizza maintains information about its customers to support their interaction with the restaurant and the processing of their orders. Customer information may include data required for identification and contact, as well as one or more addresses associated with a customer.

Food safety is particularly relevant in the restaurant business. Customers may provide information concerning known food intolerances or other relevant dietary restrictions. Such information can support informed choices and assist the restaurant in identifying potential incompatibilities between a customer's dietary restrictions and the food being ordered.

Customer information constitutes personal data and is expected to be handled in accordance with applicable data-protection requirements. Where information concerning marketing preferences or consent is collected, it should be managed explicitly and appropriately.

II.3.1.2 Products and Allergens

LaPrizza maintains a catalogue of the products relevant to its business operation. A product represents an item handled by the restaurant and may correspond, for example, to a beverage, a prepared food item, or a component used in the preparation of a pizza.

Products may be organized into categories and may have different roles within the restaurant's operation. In particular, a product may be made available for direct sale to customers, may be used as a component in the preparation of pizzas, or may fulfil both purposes.

Products may contain or be associated with one or more *allergens*. The restaurant therefore maintains information about relevant allergens and their association with products. This information provides the basis for informing customers and restaurant staff about potential food-safety concerns.

Products may also have associated commercial and operational information according to their role in the system. Sellable products may have an associated selling price, while the commercial price of a pizza may additionally depend on its selected size and configuration.

Products designated as pizza components may have size-dependent prices that provide the information required to determine the effect of individual components on the price of a configured pizza. The specific rules used to determine the final price of predefined and personalized pizzas—including the relationship between the price of the predefined pizza and the prices of added, removed, or replaced components—are established by the corresponding User Stories.

Where appropriate, dietary restrictions recorded for a customer may be related to the allergen information maintained by the system. This provides a basis for identifying potential conflicts between a customer's recorded dietary restrictions and the allergens associated with products or configured pizzas. In the case of a configured pizza, the relevant allergen information may depend on the particular products selected as its components.

Product Concept

Within the **LaPrizza** domain, a **Product** represents an item managed by the restaurant. Being a product does not imply that the item is necessarily sold directly to customers. A product may be sellable, may be used as a pizza component, may be subject to stock control, or may combine several of these characteristics.

II.3.1.3 Pizzas, Sizes, and Components

Pizzas constitute a particular type of product offered by **LaPrizza** and may be available in different predefined sizes. A *Pizza Size* characterizes the commercial size of a pizza and may influence its price and the price of the components used in its preparation.

Some products in the **LaPrizza** catalogue may be designated as *Pizza Components*. A pizza component is therefore not a distinct kind of product, but rather a product that has been identified as suitable for use in the configuration and preparation of pizzas.

Pizza components are classified according to the role they play in a pizza. Relevant component types may include, among others:

- **Base Dough** – products that may be used as the dough forming the base of a pizza;
- **Base Sauce** – products that may be used as the sauce applied to the pizza base;
- **Cheese** – products representing the cheese component of a pizza;
- **Topping** – additional products that may be added to a pizza.

These component types represent **business classifications rather than distinct kinds of products**. A product designated as a pizza component is associated with a *Pizza Component Type* that identifies the role it may play in the configuration of a pizza.

For example, *Traditional Dough* and *Wholemeal Dough* may be classified as *Base Dough*, while *Tomato Sauce* may be classified as *Base Sauce*, *Mozzarella* as *Cheese*, and products such

as *Ham*, *Mushrooms*, or *Olives* as *Toppings*.

Pizza component types should be regarded as configurable business information. The set of available types may therefore evolve without requiring changes to the fundamental product model. This also allows products belonging to the same component type to be treated consistently during pizza configuration while preserving their individual characteristics, allergen information, and pricing.

When a product is designated for use as a pizza component, it is associated with exactly one *Pizza Component Type*, which characterizes the role played by that product in pizza configuration and preparation. This designation does not create a separate product or component entity; rather, it adds pizza-specific classification and configuration information to the existing product.

Pizza Component Types may also provide the basis for defining configuration rules. Such rules may, for example, constrain the number of components of a given type that can participate in a pizza configuration or determine whether a particular type is mandatory or optional. The specific configuration rules applicable to the prototype are established by the corresponding User Stories.

The price of a pizza component may depend on the selected pizza size. Consequently, the same component may have different prices when used in pizzas of different sizes. For example, adding a particular topping to a small pizza may have a different price from adding the same topping to a large pizza.

Pizza sizes and component types are independent concepts. A pizza size establishes the size of the pizza being configured, while a component type classifies the role of a product within that configuration. Together, these concepts provide the basis for determining which products can participate in a pizza configuration and the price applicable to each selected component.

The size-dependent characteristics of a pizza component can be represented through a *Component Size Configuration*. Such a configuration associates a product designated as a pizza component with a particular *Pizza Size* and establishes the information applicable when that component is used in a pizza of that size.

For example, a Component Size Configuration may establish the price of the component and the number of abstract stock units consumed when it is used in a pizza of the corresponding size. A pizza component may therefore have different configurations for different pizza sizes.

LaPrizza may offer predefined pizzas in addition to allowing customers to create personalized configurations. A predefined pizza is represented as a sellable product and defines an initial selection of pizza components. A *Margherita*, for example, may establish a particular base dough, base sauce, cheese, and set of toppings as its default configuration.

The configuration of a predefined pizza establishes the products that form its default composition without changing their identity as independent products. This configuration can subsequently serve as the starting point for customer customization.

Within this specification, a *personalized pizza* refers to a pizza configuration created in the context of a customer interaction or order, rather than to a separate kind of catalogue product. Such a configuration may retain a reference to the predefined pizza from which it originated, together with the selected size and the component configuration resulting from the customization.

A personalized pizza may therefore be created by selecting a pizza size and modifying the components of a predefined pizza, for example by adding, removing, or replacing available components. The resulting configuration preserves the individual products selected by the customer,

allowing its price, relevant allergen information, stock consumption, and preparation requirements to be determined from the resulting configuration.

The allergen information of a pizza can be derived from the products participating in its current configuration. Consequently, customization may change the allergens associated with the resulting pizza as components are added, removed, or replaced.

The combination of *Pizza Sizes*, *Pizza Component Types*, and products designated as *Pizza Components* thus provides a flexible basis for representing both predefined and customer-configured pizzas without introducing separate product types for each category of pizza component.

II.3.1.4 Stock Control

Some products may additionally be subject to *stock control*. For these products, **LaPrizza** maintains information about the quantity currently available. Whether stock control is applicable depends on the nature and operational role of the product. For example, beverages and products used as pizza components may represent physical items whose available quantities need to be monitored, whereas a pizza prepared on demand does not necessarily represent an independently stocked item.

For the purposes of the prototype, stock is represented using *abstract stock units*. A stock unit represents a normalized quantity of a product used for stock-control purposes, without requiring the prototype to model physical measurement units such as kilograms, litres, packages, or individual portions. The meaning and granularity of one stock unit are product-specific and are assumed to have been previously established by **LaPrizza**.

This abstraction allows stock quantities and consumption values to be represented as non-negative integer numbers of units. Replenishment increases the number of units available, whereas the sale or operational use of a stock-controlled product reduces it according to the applicable consumption value. The business event at which the corresponding stock reduction is recorded depends on the operational workflow established by the applicable User Stories.

For directly sellable products, such as beverages, stock consumption may be determined from the quantity sold and the number of stock units associated with each sold item. Products used as pizza components require an additional consideration: the quantity consumed may depend on the size of the pizza in which the component is used. This provides a simplified way of representing, for example, that preparing a large pizza may consume more mozzarella than preparing a small pizza without requiring the system to represent the corresponding physical quantities.

A product being designated as sellable indicates that it may be offered to customers; it does not necessarily imply that it is currently available for ordering. Availability may additionally depend on operational conditions, including the available stock of the product itself or, where applicable, of products required for its preparation.

For a product designated as a pizza component, the applicable stock consumption may depend on the selected pizza size. This information is established by the corresponding *Component Size Configuration*, together with the size-dependent price described previously.

Table 1 illustrates this concept. The values are merely illustrative and do not represent physical measurement units.

This deliberately simplified stock model allows the prototype to reason about product availability and consumption while avoiding the complexity of physical units of measurement, packaging, portioning, and unit conversion. It may also provide information that can subsequently

Table 1: Example of size-dependent pricing and stock consumption for pizza components.

Pizza Component	Pizza Size	Price	Stock Units Consumed
Mozzarella	Small	1.00 €	1
Mozzarella	Medium	1.50 €	2
Mozzarella	Large	2.00 €	3
Ham	Small	1.00 €	1
Ham	Medium	1.50 €	1
Ham	Large	2.00 €	2

support other operational concerns, such as identifying products that require replenishment.

II.3.1.5 Customer Orders

Customers interact with **LaPrizza** primarily by placing orders. An order consists of one or more *Order Items*, each representing something requested by the customer together with the information required to determine its quantity, price, and subsequent processing.

An order item may correspond directly to a sellable product from the **LaPrizza** catalogue. In the case of a personalized pizza, however, the order item may instead preserve the particular pizza configuration selected by the customer. This distinction allows personalized pizzas to participate in an order without requiring each resulting configuration to become a new catalogue product.

Before confirming an order, the system may allow its contents and price to be simulated, enabling the customer to review an intended purchase without creating an effective order.

For a personalized pizza, the corresponding order item preserves sufficient information about the selected predefined pizza, size, and component configuration to represent what was effectively ordered. This preserved configuration provides the basis for determining its price, allergens, stock consumption, and preparation requirements independently of subsequent changes to the corresponding catalogue information.

Once confirmed, an order may progress through different stages of the restaurant's operational process. Since an order can contain several items that may be prepared and served at different moments, its state may also reflect the progress of individual items. This allows the restaurant to represent, where appropriate, which items are awaiting preparation, being prepared, ready to be served, or already served. The overall state of an order may consequently be derived from, or coordinated with, the state of its constituent items.

II.3.1.6 Food Preparation and Operational Planning

Confirmed customer orders may generate preparation work to be performed by **LaPrizza's** kitchen. Depending on the item ordered, fulfilling an order item may involve one or more *Preparation Activities*. For pizzas, a simplified preparation process may, for example, include activities associated with assembly and baking.

Products requiring preparation may provide the operational information from which these activities are determined. Such information may describe the expected duration of an activity, the type of resource required to perform it, and dependencies on other activities. When an order is processed, this information provides the basis for determining the preparation work associated with the corresponding order items.

Preparation activities may depend on the completion of other activities, thereby establishing sequencing constraints within the preparation process. They may also require access to limited *Kitchen Resources*, such as preparation stations or ovens.

Kitchen resources may represent individual resources or groups of equivalent resources, depending on the level of detail required by the scheduling problem. Relevant characteristics may include their availability and capacity. Ovens are a particularly important example, since their capacity and availability directly influence the scheduling of pizza preparation.

Coordinating preparation work from multiple orders therefore involves considering the activities to be performed, their expected durations and dependencies, and the availability and capacity of the required kitchen resources. Different scheduling approaches may be used to organize this work with the objective of completing orders efficiently while making appropriate use of the available resources.

A *food-preparation schedule* can represent how pending activities are sequenced and, where applicable, allocated to kitchen resources. As restaurant activity evolves and new orders are received, or as resource availability changes, such a schedule may need to be revised to reflect the current operational situation.

This planning information can also provide the basis for an interactive representation of kitchen operations, allowing restaurant staff to visualize planned and ongoing preparation activities and the utilization of relevant resources.

II.3.1.7 Conceptual Domain Overview

The business concepts introduced in the preceding sections form a deliberately simplified representation of **LaPrizza**'s domain. Their relationships provide a common conceptual basis for the different functional and technological areas explored throughout the project.

Figure 1 highlights the concepts most directly related to products, pizza configuration, orders, stock, and food preparation. It is intended as a conceptual overview rather than as a prescriptive domain model: it does not represent every concept described in this specification and should not be interpreted as defining the classes, entities, aggregates, database structures, or other implementation artefacts that teams are required to adopt.

Core Domain Concepts

The **LaPrizza** prototype adopts a deliberately simplified product, pizza configuration, stock, and operational model:

- a **Product** represents any item managed by the restaurant;
- a product may be **sellable**, **stock-controlled**, designated as a **Pizza Component**, or combine these roles;
- a **Pizza Component Type** classifies the role played by a component in a pizza

configuration;

- a **Pizza Size** characterizes the commercial size of a pizza and may influence both pricing and component consumption;
- a **Component Size Configuration** establishes the size-dependent information applicable to a pizza component, such as its price and the number of abstract stock units consumed;
- a **Predefined Pizza** is represented by a sellable product associated with a default pizza configuration;
- a **Personalized Pizza** refers to an order-specific **Pizza Configuration**, typically derived from a predefined pizza, and does not constitute a separate catalogue product;
- stock is represented using **abstract stock units**, avoiding the need to model physical units of measurement and conversions;
- customer orders contain **Order Items**, which may correspond directly to catalogue products or preserve personalized pizza configurations;
- fulfilling an order item may give rise to one or more **Preparation Activities**; the activities required for an item may be determined from operational information associated with the ordered product and, where applicable, its pizza configuration;
- Preparation Activities may depend on other activities and may require available **Kitchen Resources**.

About the Conceptual Diagram

The diagram presented in Figure 1, provides a **simplified, non-technical, and non-exhaustive representation** of selected concepts and relationships within the **LaPrizza** business domain. Its purpose is to support the understanding of the system specification and to provide a common conceptual view of the problem domain.

It must **not be interpreted as a formal domain, data, or software design model, nor does it prescribe classes, entities, aggregates, attributes, cardinalities, database structures, or implementation decisions**.

The diagram does not replace the appropriate modelling formalisms, techniques, and artefacts that teams are expected to apply during the analysis and design of the system. Teams remain responsible for analysing the specification and developing their own technical models according to the principles and practices addressed by the corresponding ??.

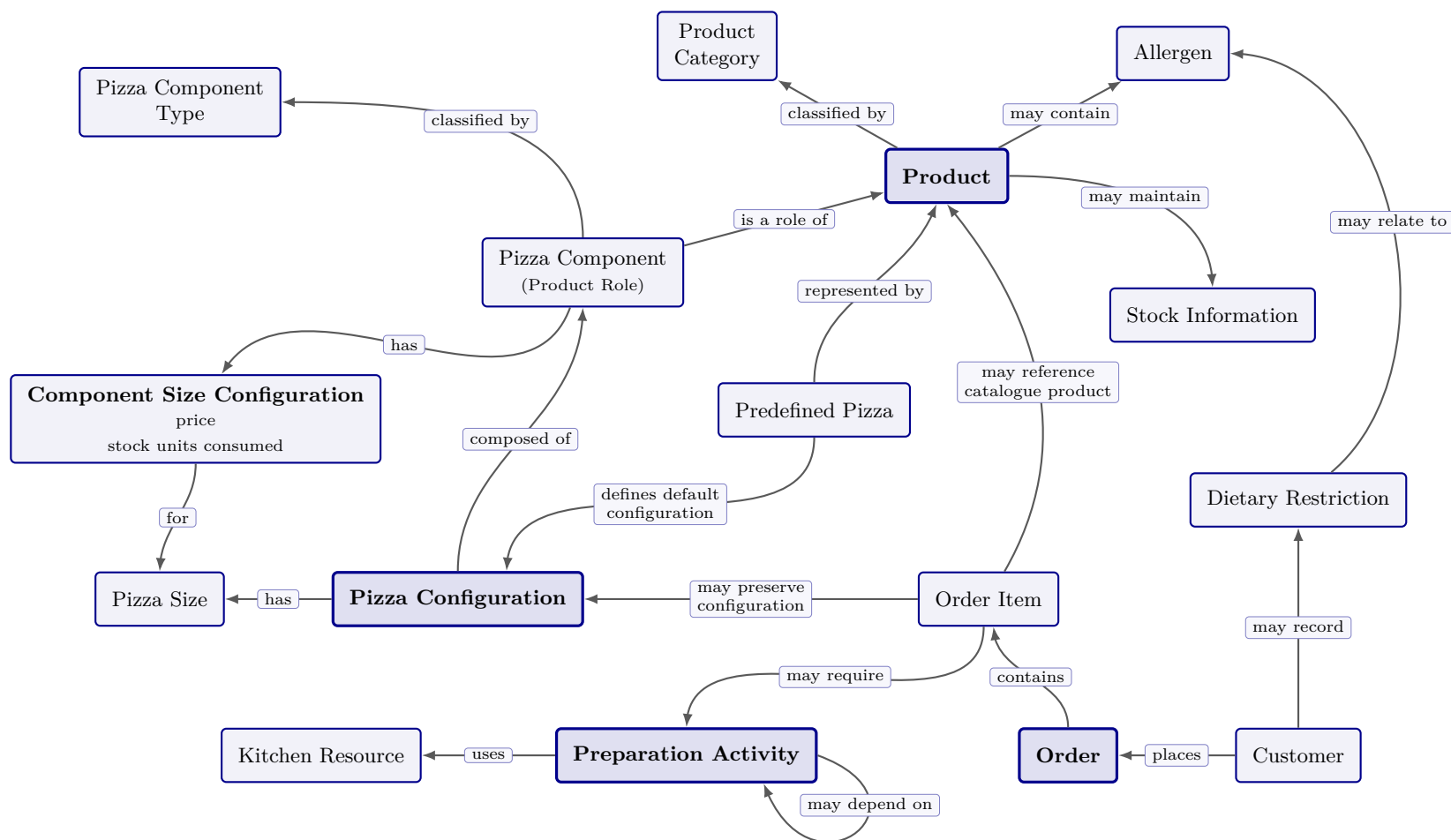


Figure 1: Conceptual overview of the main **LaPrizza** business domain concepts.

II.3.2 Web Application

The Web Application is envisioned as the main interface to the **LaPrizza** information system and as the primary means of accessing the business capabilities associated with the domain concepts introduced in the preceding Business Overview.

Depending on the functionality introduced throughout the project, the Web Application may provide support for business areas such as:

- **Identity and Access Management:** identification and authentication of users and control of access to functionality according to their assigned roles and permissions;
- **Customer Management:** registration, consultation, and management of customer information, including relevant dietary information;
- **Product and Allergen Management:** management of products, product categories, applicable allergen information, and relevant commercial and operational properties;
- **Stock Control:** representation and monitoring of current stock levels and consumption information for products subject to stock control;
- **Pizza Configuration:** management of pizza sizes, pizza component types, products available as pizza components, their size-dependent prices, and predefined pizza configurations;
- **Order Management:** creation and management of customer orders, including personalized pizzas, and their progression through the restaurant's operational process;
- **Food Preparation Management:** consultation and, where applicable, management of preparation activities, kitchen resources, and the operational state of ordered items.

As these capabilities are introduced, they should be made available through a coherent user interface and, where integration with other modules is required, through appropriate interfaces or APIs.

Particular attention should be given to allergen information whenever functionality involving products, pizza configurations, or customer orders is implemented. Since the allergens associated with a pizza may originate from any of its selected components, the available information should enable customers and **LaPrizza** staff to identify the relevant allergens for both predefined and personalized pizzas.

II.3.3 3D Visualization and Interaction

Interactive 3D visualization constitutes one of the technological areas explored within the **LaPrizza** information system. The corresponding web-based modules are expected to use WebGL and *Three.js* and to interact, where appropriate, with information maintained by the remaining system rather than operating exclusively as isolated static representations.

The project may explore different visualization and interaction scenarios, including the following.

II.3.3.1 Build Your Pizza

The *Build Your Pizza* scenario envisions an interactive 3D interface through which customers can visualize and configure a personalized pizza.

The interaction may allow users to select a pizza size and choose among the products available as pizza components. Components may be presented according to their role in the pizza, including base doughs, base sauces, cheeses, and toppings.

A customer may start from a predefined pizza, such as a *Margherita*, and interactively modify its composition by adding, removing, or replacing permitted components. The three-dimensional representation should evolve accordingly, providing visual feedback about the resulting configuration.

Where this scenario is implemented, the visualization is expected to obtain information concerning available pizza sizes, components, component types, applicable configuration rules, prices, and other relevant product information through interfaces exposed by the **LaPrizza** information system.

A resulting pizza configuration provides the information required for integration with the ordering functionality, allowing the selected size, components, calculated price, and relevant allergen information to be associated with a customer order.

II.3.3.2 Kitchen and Food Preparation Visualization

The *Kitchen and Food Preparation Visualization* scenario envisions an interactive 3D representation of **LaPrizza**'s kitchen operations, allowing users to observe a schedule for the preparation of customer orders and its evolution over time.

The visualization may represent relevant kitchen resources and show how food-preparation activities are scheduled and progress through the preparation process. For pizzas, this may include their preparation, allocation to available ovens, expected baking sequence and timing, and the utilization of the corresponding kitchen resources.

Where integrated with the Artificial Intelligence module, scheduling information can be obtained from the schedule produced by that module, providing consistency between the planned activities and their visual representation.

The visualization may distinguish activities that are waiting, scheduled, currently being performed, or completed. It may additionally represent timing information, resource availability, dependencies between preparation activities, and potential scheduling conflicts or delays.

This scenario therefore provides an opportunity to explore a spatial and temporal representation of **LaPrizza**'s food-preparation process, complementing conventional operational interfaces with a visual perspective on kitchen scheduling and resource utilization.

II.3.4 Artificial Intelligence and Food Preparation Scheduling

Efficient operation of **LaPrizza** involves preparing customer orders in a timely and coordinated manner while making appropriate use of available kitchen resources. Since different food items may involve different preparation activities, durations, resources, and sequencing constraints, food preparation provides a suitable scheduling and decision-support problem for the application of Artificial Intelligence techniques.

Within this context, the Artificial Intelligence component is intended to explore the **planning and scheduling of food preparation activities**. Given a set of orders awaiting preparation, the problem consists of determining how the corresponding activities can be organized and allocated to available kitchen resources.

The scheduling problem may consider information maintained by the **LaPrizza** information system, including the order items awaiting preparation, the preparation activities required to fulfil them, their estimated processing times and dependencies, and the availability and capacity of the corresponding kitchen resources. For pizzas, this may include activities associated with assembly and baking, together with the allocation and sequencing of baking activities using the available ovens.

For the purposes of the **IART** contribution to the Integrative Project, the planning and reasoning mechanisms are to be developed using *Prolog*, applying Artificial Intelligence algorithms and techniques appropriate to the scheduling problem addressed in the corresponding Sprint Backlogs.

A resulting food-preparation schedule may identify the sequence and timing of activities and, where applicable, the kitchen resources assigned to them. Such information can subsequently be made available to other components of the system, including conventional Web Application interfaces and the *Kitchen and Food Preparation Visualization* module.

Different scheduling strategies and optimization criteria may progressively be explored throughout the project. Possible operational objectives include reducing customer waiting times, meeting expected order completion times, minimizing idle resources, or improving the overall utilization of the kitchen.

II.3.5 Systems Administration and Infrastructure

The **LaPrizza** information system provides a context in which appropriate infrastructure and systems-administration practices can be explored to support the deployment and operation of its different components in a secure, reproducible, reliable, and maintainable manner.

As the prototype evolves, systems-administration activities may address the deployment and operation of the developed solution throughout its lifecycle. The infrastructure should progressively support automated and repeatable deployment procedures and appropriate isolation between system components and deployment environments.

Relevant concerns include access control and infrastructure security. Administrative access to underlying systems should be appropriately restricted and monitored, while relevant security and operational events may be logged to support auditing and incident analysis.

The protection and preservation of **LaPrizza's** data should also be considered. Appropriate backup mechanisms and retention policies may be defined and automated to support the recovery of relevant information following accidental deletion, data corruption, infrastructure failure, or other representative incidents.

Business continuity and disaster recovery provide additional concerns that may be considered as the infrastructure evolves. This may involve identifying critical system components and associated risks, establishing suitable recovery objectives and procedures, and evaluating the ability of the proposed infrastructure to restore relevant services and data following representative failure scenarios.

The infrastructure design may additionally consider availability and scalability. Depending

on the requirements introduced during the project, mechanisms such as reverse proxies, load balancing, service replication, containerization, or equivalent approaches may be employed and evaluated.

Overall, this area provides an opportunity to demonstrate how systems-administration practices contribute to the security, availability, recoverability, reproducibility, and maintainability of the integrated **LaPrizza** information system.

II.3.6 Integration and General System Concerns

Although the proposed solution is divided into different functional and technological areas, these areas are intended to contribute to a **single integrated information system** supporting **LaPrizza**'s business and operational activities. The business concepts introduced in the Business Overview provide a common domain context for this integration.

As functionality is progressively introduced, the Web Application may provide access to the system's business capabilities, while the 3D Visualization and Artificial Intelligence components may consume or produce information concerning the same underlying business concepts. Their integration therefore provides an opportunity to define and document the high-level system architecture, the responsibilities and boundaries of each component, data ownership, and the interfaces or APIs used for communication.

Security and data protection constitute cross-cutting concerns and should be considered wherever applicable. Personal data, particularly customer information, is expected to be handled in accordance with applicable data-protection requirements, including the General Data Protection Regulation (GDPR). Authentication, authorization, access control, and other data-protection mechanisms should therefore be incorporated where required by the functionality implemented.

The **LaPrizza** information system should also be designed with qualities such as performance, maintainability, and scalability in mind. The resulting prototype is expected to demonstrate not only the technologies and techniques associated with the different project areas but, more importantly, their progressive integration into a coherent solution addressing a realistic business problem.